AWS EKS Cluster Configuration, Setup & App Deployment

This documentation provides detailed instructions for setting up and configuring an Amazon Elastic Kubernetes Service (EKS) cluster in the **ap-south-1** region. The steps include installing AWS CLI, setting up the EKS cluster, updating kubeconfig, installing kubectl, installing eksctl, associating IAM OIDC provider, creating a service account with required permissions, deploying the AWS EBS CSI Driver, and applying a custom manifest file.

Step 1: Install AWS CLI

Follow these steps to install the AWS Command Line Interface (CLI):

1. Download the AWS CLI installer:

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

2. Install the unzip utility (if not already installed):

```
sudo apt install unzip
```

3. Unzip the installer:

```
unzip awscliv2.zip
```

4. Run the installer:

```
sudo ./aws/install
```

5. Configure AWS CLI:

```
aws configure
```

Step 2: Clone the Terraform Project and Deploy EKS

1. Clone the Terraform project folder and navigate to it:

   git clone <terraform_project_repo>

cd <terraform_project_folder>

2. Initialize Terraform:

terraform init

3. Deploy the resources:

terraform apply --auto-approve

Step 3: Update kubeconfig for the EKS Cluster

Run the following command to update the kubeconfig file with your EKS cluster details:

aws eks --region ap-south-1 update-kubeconfig --name devopsshack-cluster

This command retrieves the kubeconfig configuration for the devopsshack-cluster and stores it in the default kubeconfig file (e.g., ~/.kube/config).

Step 4: Install kubectl

Follow these steps to install the latest version of kubectl:

1. Download the kubectl binary:

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

2. Make the binary executable:

```
chmod +x kubectl
```

3. Move the binary to a system PATH directory:

```
sudo mv kubectl /usr/local/bin/
```

4. Verify the installation:

```
kubectl version --client
```

Step 5: Install eksctl

Follow these steps to install eksctl, a CLI tool for managing EKS clusters:

1. Download the eksctl tarball:

```
curl -LO "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.gz"
```

2. Extract the tarball:

```
tar -xzf eksctl_Linux_amd64.tar.gz
```

3. Move the binary to a system PATH directory:

```
sudo mv eksctl /usr/local/bin
```

4. Verify the installation:

```
eksctl version
```



Step 6: Associate IAM OIDC Provider

Associate an IAM OIDC provider with your EKS cluster to enable IAM roles for Kubernetes service accounts:

```
eksctl utils associate-iam-oidc-provider --region ap-south-1 --cluster devopsshack-cluster --approve
```

This command associates the OIDC provider required for enabling IAM roles for service accounts in the EKS cluster.

Step 7: Create an IAM Service Account

Create a Kubernetes service account with IAM permissions for the Amazon EBS CSI Driver:

```
eksctl create iamserviceaccount \
  --region ap-south-1 \
  --name ebs-csi-controller-sa \
  --namespace kube-system \
  --cluster devopsshack-cluster \
  --attach-policy-arn arn:aws:iam::aws:policy/service-role/AmazonEBSCSIDriverPolicy \
  --approve \
  --override-existing-serviceaccounts
```

- --name ebs-csi-controller-sa: Name of the service account.
- --namespace kube-system: Namespace where the service account will be created.
- --attach-policy-arn: IAM policy ARN for EBS CSI Driver permissions.
- --approve: Automatically approve the creation.

Step 8: Deploy the AWS EBS CSI Driver

Deploy the AWS EBS CSI Driver in the cluster using the following command:

```
kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable/ecr/?ref=release-1.11"
```

This command deploys the latest stable release of the EBS CSI Driver from the official repository.

Step 9: Deploy Application Using Manifest File

After setting up the cluster, deploy your application using a Kubernetes manifest file:

```
kubectl apply -f manifest.yaml
```

Replace manifest.yaml with the path to your Kubernetes manifest file.

Verification

1. Verify that the kubeconfig is updated and the cluster is accessible:

```
kubectl get nodes
```

2. Verify that the service account is created:

```
kubectl get serviceaccount ebs-csi-controller-sa -n kube-system
```

3. Verify the deployment of the EBS CSI Driver:

```
kubectl get pods -n kube-system -l app.kubernetes.io/name=aws-ebs-csi-driver
```

4. Verify the application deployment:

```
kubectl get pods
```
